25.2.6 In general, mechanisms that allows consistency in the complete INS with regards to alert management.

26 Integration of systems in alert management

26.1.1 All systems, sources and sensors incorporated, connected in the INS should be part of the alert management.

26.1.2 The following equipment and systems, if installed, and not inco rporated in the INS should be also included in the alert management as far as possible:

- heading information system
- heading/track control system
- electronic position-fixing systems
- speed and distance measuring equipment
- radar with target tracking functions
- ECDIS
- AIS
- echo sounding equipment
- GMDSS equipment
- relevant machinery alarms for early warning.

26.1.3 The following equipment and systems, if installed, should be connected to the alert management:

- bridge navigational watch alarm.

Module D – Documentation requirements

27 Manuals

27.1 Operating manuals should include:

- an overall functional description of the INS
- the redundancy concept and the availability of functions

---CHUNK---

SOURCE: MSC.252(83).pdf
- bridge navigational watch alarm.

Module D – Documentation requirements

27 Manuals

27.1 Operating manuals should include:

- an overall functional description of the INS
- the redundancy concept and the availability of functions
- a description of possible failures and their effects on the system (e.g. by using part of the failure analysis)
- guidance for the adjustment of the limits for alerts
- the implications of using different reference locations

details of the mechanism for marking valid, doubtful and invalid data
for an INS providing automatic control functions (e.g. for heading, track or speed) details of the external override and/or bypassing devices used in the reversionary mode.

---CHUNK---

SOURCE: MSC.252(83).pdf
for an INS providing automatic control functions (e.g. for heading, track or speed) details of the external override and/or bypassing devices used in the reversionary mode.

27.2 The installation manuals should include adeq uate information to allow the INS to be installed so that it can meet all requirements adopted by the Organization.

27.3 The installation manuals should include the following:

details of sources, components and the interconnections forming the INS
details of the interfaces and connections for data import and export and the interconnection diagrams and interfacing details for extern al parts of the INS and for devices, sensors to be connected
instructions for the installation a nd connection of facilities for alert acknowledgement and cancellation including the back-up officer alarm in case of an INS providing automatic control functions (e.g. for heading, track or speed)
the details of the power supply arrangements
recommendations on the physical layout of equipment and necessary space for maintenance
for an INS providing automatic control functions (e.g. for heading, track or speed)

---CHUNK---

SOURCE: MSC.252(83).pdf
the details of the power supply arrangements
recommendations on the physical layout of equipment and necessary space for maintenance
for an INS providing automatic control functions (e.g. for heading, track or speed)
details of the installation and connection of external overri de and/or bypassing devices used in the reversionary mode and if rudder angle, heading, propulsion data – e.g. power, propeller pitch, are not be presented on a display of the INS workstation, the necessary details.

28 Information regarding the system configuration

28.1 Manufacturer or system in tegrator of INS should declare the following information relating to the system configuration, if applicable:

- basic system configuration
- interconnecting block diagram (Hardware)
- sources identification
- override
- priority of control (task stations)
- data flow schematic diagram and its interpretation
- default conditions
- back-up arrangement
- redundancy arrangement
- explanation of scope to fulfil requirements of SOLAS regulation V/19 with particular INS (for one equipment concept)

---CHUNK---

SOURCE: MSC.252(83).pdf
- default conditions
- back-up arrangement
- redundancy arrangement
- explanation of scope to fulfil requirements of SOLAS regulation V/19 with particular INS (for one equipment concept)

other useful materials for inspector (such evidence of fulfilled requirements as other means).

29 Failure analysis

29.1.1 A failure analysis, at INS functional level, should be performed and documented for the INS. The failure analysis should verify that th e INS is designed on “fail-to-safe” principle and that failure of one part of the integrated system should not affect the functionality of other parts, except for those functions directly dependent on the defective part.

30 Guidance to equipment manufacturers for th e provision of onboard familiarization material

Material enabling onboard familia rization training should be pr ovided for the INS. The onboard familiarization material should explain all c onfiguration, functions, limitations, controls,

---CHUNK---

SOURCE: MSC.252(83).pdf
material

Material enabling onboard familia rization training should be pr ovided for the INS. The onboard familiarization material should explain all c onfiguration, functions, limitations, controls,

displays, alerts and indications of the INS. Guidance and recommendations to the equipment manufacturers for the provision of onboard familiarization material are given in Appendix 2.

Appendix 1

DEFINITIONS

Added Value The functionality and information, which are provided by the INS, in addition to the requirements of the performance standard for the individual equipment.

Alarm An alarm is the highest priority of an alert. Condition requiring immediate attention and action by the bridge